1..10

1	2	3	4	5	6	7	8	9	10
a	b	a	d	b	b	a	a	c	a

11. a) $L = 1000\,B = 8000\,b$

$R_{LAN} = 10\,Mb/s = 10^7\,b/s$

$v = 2 \times 10^8\,m/s$

$l_{LAN} = 20\,m$

$$d_{trans.LAN} = \frac{L}{R_{LAN}} = \frac{8000}{10^7} = \frac{8 \times 10^3}{10^7} = 8 \times 10^{-4}\,s$$

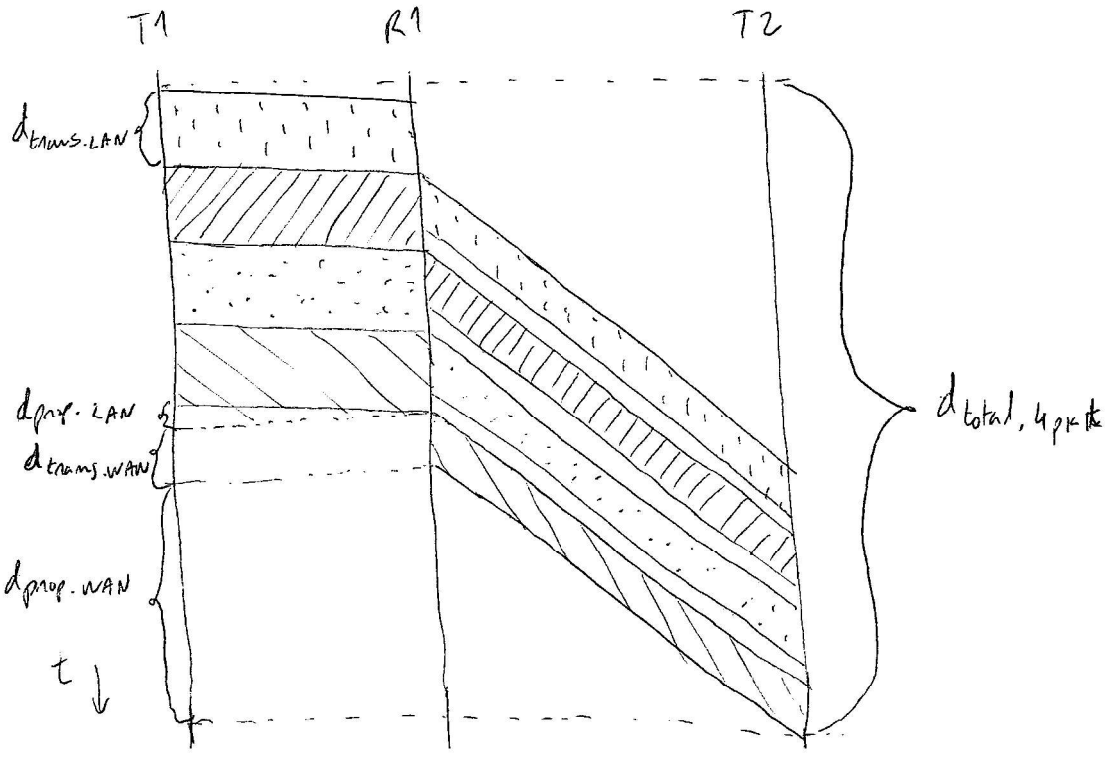
$$d_{prop.LAN} = \frac{l_{LAN}}{v} = \frac{20}{2 \times 10^8} = \frac{2 \times 10^1}{2 \times 10^8} = 10^{-7}\,s$$

b) $d_{trans.WAN} = 0{,}6\,ms = 6 \times 10^{-4}\,s$

$d_{prop.WAN} = 15\,ms = 1{,}5 \times 10^{-2}\,s$

$$d_{total,1pkt} = d_{trans.LAN} + d_{prop.LAN} + d_{trans.WAN} + d_{prop.WAN}$$
$$= 8 \times 10^{-4} + 10^{-7} + 6 \times 10^{-4} + 1{,}5 \times 10^{-2}$$
$$= 1{,}64001 \times 10^{-2}\,s \quad (\simeq 16{,}4\,ms)$$

c)



d) $d_{total.4pkt} = 4d_{trans.LAN} + d_{prop.LAN} + d_{trans.WAN} + d_{prop.WAN}$

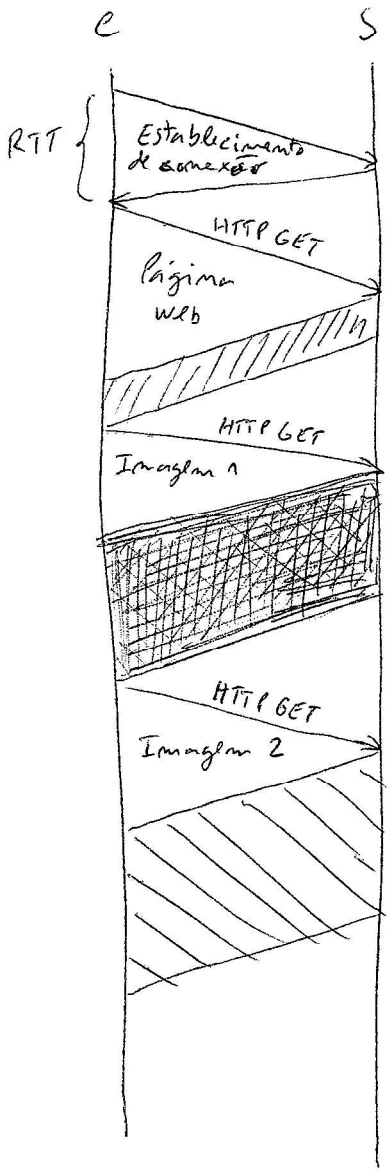
$= 4 \times 8 \times 10^{-4} + 10^{-7} + 6 \times 10^{-4} + 1,5 \times 10^{-2}$

$= 3,2 \times 10^{-3} + 10^{-7} + 6 \times 10^{-4} + 1,5 \times 10^{-2}$

$= 1,88001 \times 10^{-2}$ s $\quad (\simeq 18,8$ ms$)$

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12. a)



b) $d_{prop} = 30$ ms $= 3 \times 10^{-2}$ s $\qquad RTT = 2 \times d_{prop} = 6 \times 10^{-2}$ s

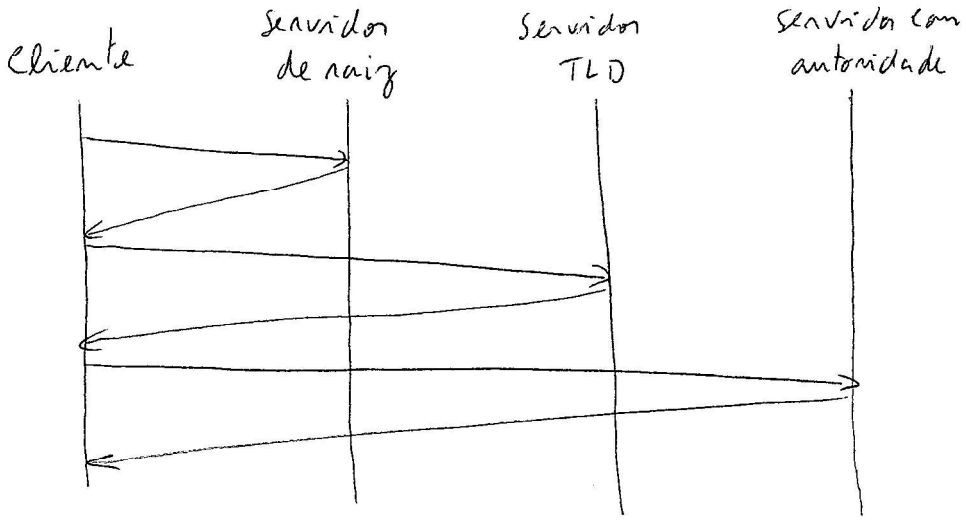
$L_{HTML} = 10$ kB $= 8 \times 10^4$ b $\qquad L_{IMG} = 200$ kB $= 1,6 \times 10^6$ b

$d_{trans.HTML} = \frac{L_{HTML}}{R} = \frac{8 \times 10^4}{8 \times 10^7} = 10^{-3}$ s $\qquad R = 80$ Mb/s $= 8 \times 10^7$ b/s

$d_{trans.IMG} = \frac{L_{IMG}}{R} = \frac{1,6 \times 10^6}{8 \times 10^7} = \frac{16 \times 10^5}{8 \times 10^7} = 2 \times 10^{-2}$ s

$d_{total} = 4 \times RTT + d_{trans.HTML} + 2 \times d_{trans.IMG} = 24 \times 10^{-2} + 10^{-3} + 4 \times 10^{-2} = 2,81 \times 10^{-1}$ s

13.


Cliente
Servidor de raiz
Servidor TLD
Servidor com autoridade